T.4 Cross Downstream Task Tests

We perform cross downstream task tests where we compare the layerwise intermediate representations between two models of the same architecture type but trained on different tasks. The two tasks we use are Node Classification and Link Prediction. The size of the intermediate embeddings is the same between the two models (Node Classification model vs Link Prediction model) except for the last layer. Since NSA is agnostic to the dimension of the embedding, we can compare all the layers regardless of their dimensionality. Just like the previous experiments, we test on the Amazon Computers Dataset and the three Planetoid Datasets. We observe that there is no strong correlation as we go further down the layers between the two models. The highest relative similarity is in the early layers and as we get to the final layers the similarity is almost non existent.

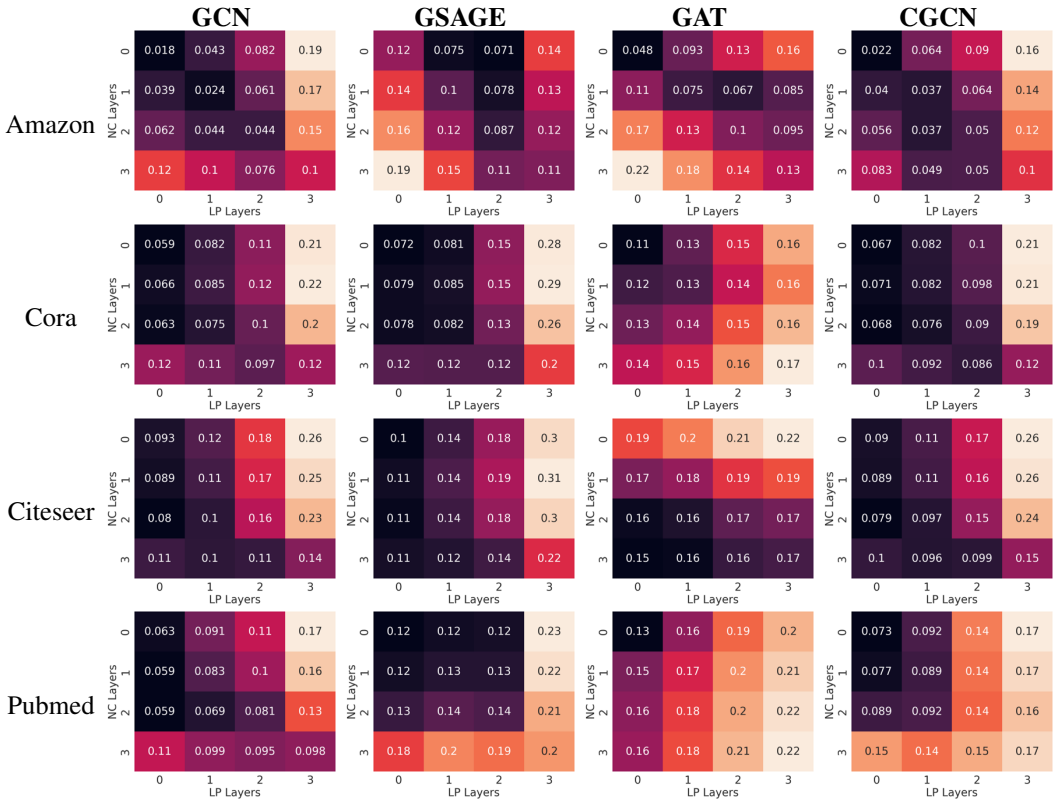
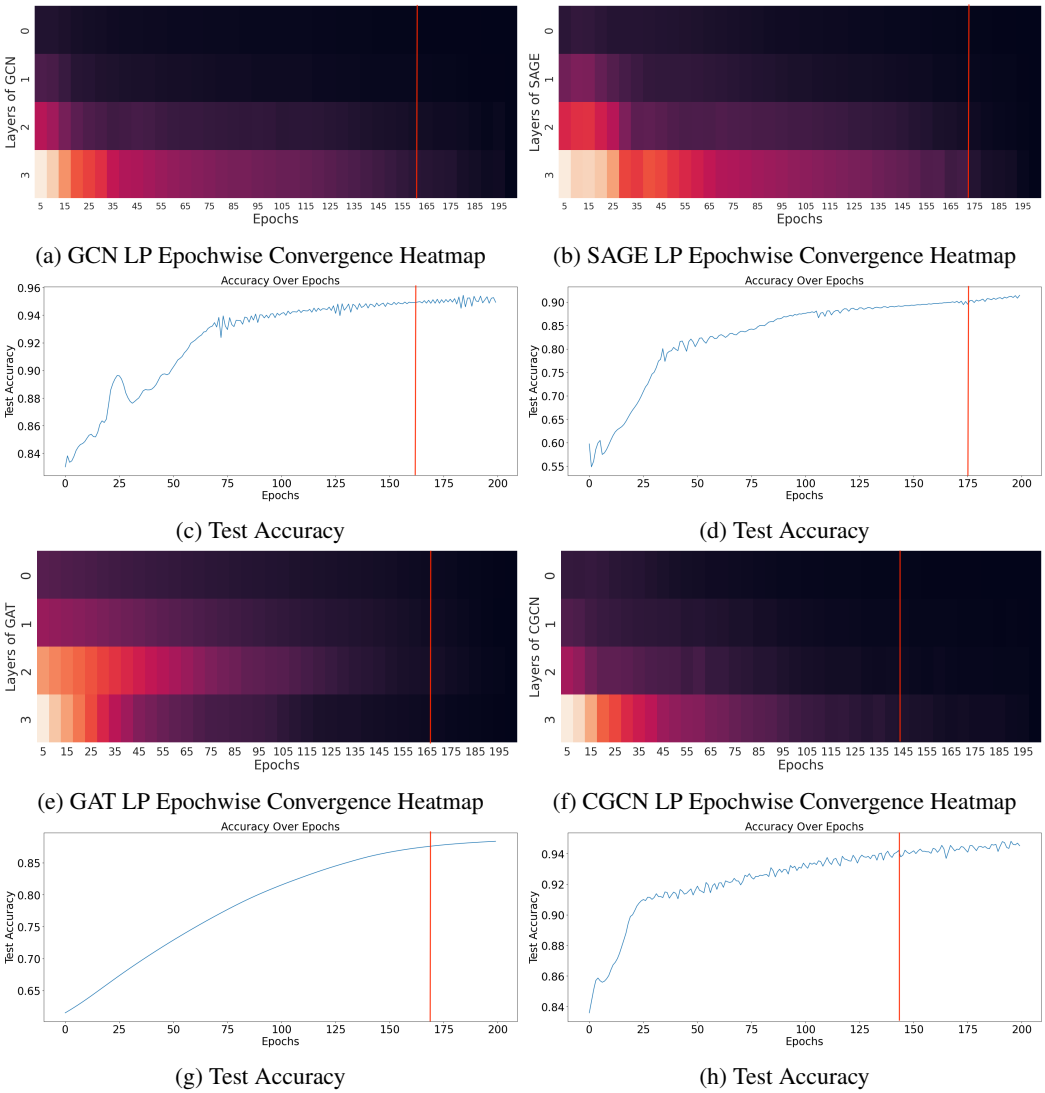


Figure 23: Cross Downstream Task Tests with NSA on the Amazon, Cora, Citeseer and Pubmed Datasets

U Convergence Tests

As a neural network starts converging, the representation structure of the data stops changing significantly and the loss stops dropping drastically. Ideally, a similarity index should be capable of capturing this change. We examine epoch-wise convergence by comparing representations between the current and final epochs. We observe that NSA starts dropping to low values as the test accuracy stops increasing significantly. This shows that NSA can be used to help approximate convergence and even potentially be used as an early stopping measure during neural network training, stopping the weight updates once the structure of your intermediate representations stabilize. We show results for all 4 architectures on node classification and link prediction. Link Prediction models were trained for 200 epochs and show similar patterns to the ones observed with node classification. All convergence tests are performed with NSA only. We show the convergence tests on all 4 architectures on Node Classification in Figure 25 and on Link Prediction in Figure 24 .



(a) GCN LP Epochwise Convergence Heatmap

(b) SAGE LP Epochwise Convergence Heatmap

(c) Test Accuracy

(d) Test Accuracy

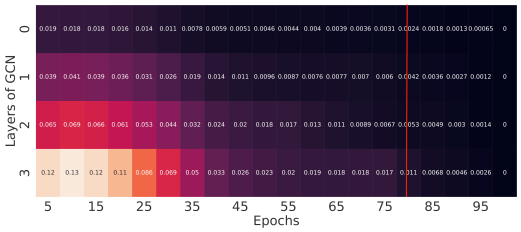
(e) GAT LP Epochwise Convergence Heatmap

(f) CGCN LP Epochwise Convergence Heatmap

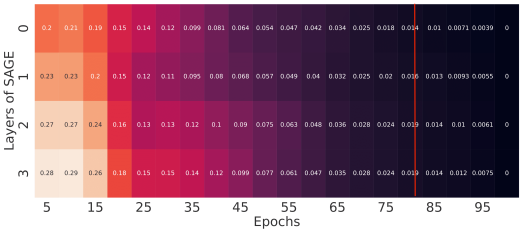
(g) Test Accuracy

(h) Test Accuracy

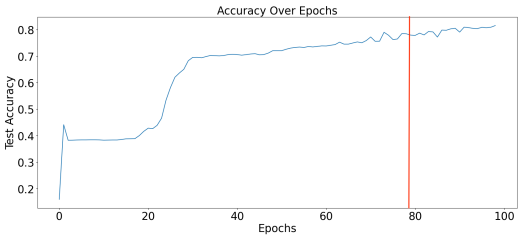
Figure 24: Convergence Tests on Link Prediction



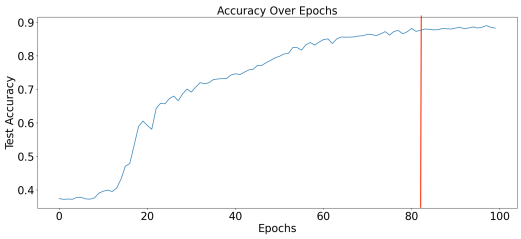
(a) GCN NC Epochwise Convergence Heatmap



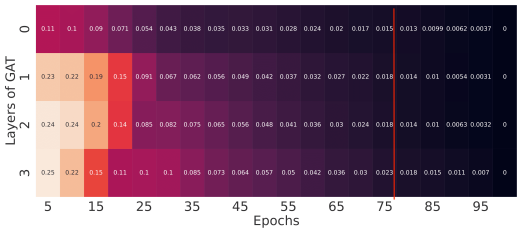
(b) SAGE NC Epochwise Convergence Heatmap



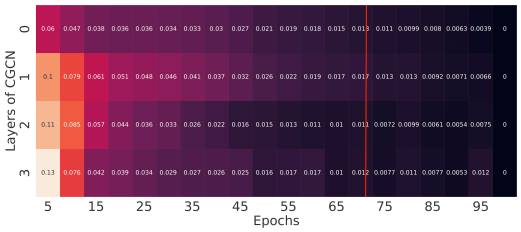
(c) Test Accuracy



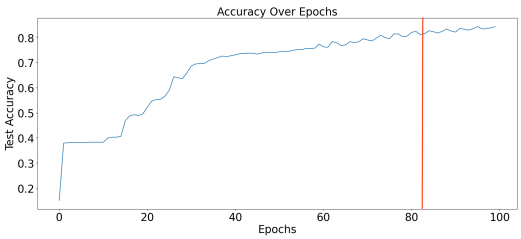
(d) Test Accuracy



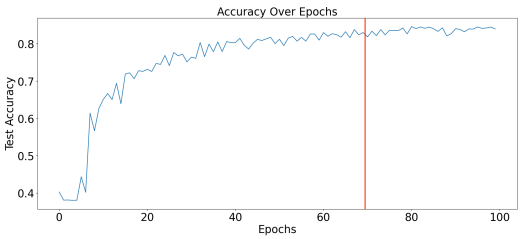
(e) GAT NC Epochwise Convergence Heatmap



(f) CGCN NC Epochwise Convergence Heatmap



(g) Test Accuracy



(h) Test Accuracy

Figure 25: Convergence Tests on Node Classification